

ChatGPT and Pi AI under the Microscope: Artificial Empathy for Enhancing Digital Customer Service

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Abstract

Digital customer service provided through artificial empathy AI chatbots, particularly when designed with empathetic characteristics, constitutes a significant factor for eliciting positive customer responses and perceptions. However, can today's consumers perceive empathy features embedded within a chatbot that serves them, and concurrently, is it feasible for an AI chatbot to offer personalized and individualized service? Many advancements in technologies such as machine learning have dramatically increased the penetration of chatbots in the digital landscape and everyday customer service interactions. AI chatbots, have considered as popular digital business tools, enhancing employee productivity, and improving customer service effectiveness. This research investigates the extent to which consumers perceive empathy in AI chatbots and explores whether the chatbots involved in this study can effectively contribute to delivering personalized service. This research investigates the extent to which consumers perceive artificial empathy in AI chatbots and explores whether the chatbots involved in this study can effectively contribute to delivering personalized service.

CCS Concepts

• Computing methodologies; • Artificial intelligence;

Keywords

Human-Computer Interaction, Artificial Intelligence, Chatbots, Artificial Empathy, Customer Service, Trust, Service Quality, Conversational Agents

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1 INTRODUCTION

Artificial empathy is when an AI system can recognize, comprehend, and copy empathetic behaviors to users without feeling anything

[7]. This simulation commonly includes affective computing frameworks that help the chatbot pick up on emotional cues and behave in ways that are comparable to how people demonstrate empathy. According to [8] there are two main types of empathy: cognitive empathy, which is knowing how someone else feels, and affective empathy, which is sharing emotional states. Even though artificial empathy is fake by nature, studies claimed that using it in the right way might make users happier, more trusting, and more aware of other people's presence [5]. But there is still disagreement over how real users feel about these kinds of responses, which makes it hard to build future chatbots. By expanding the idea of artificial empathy, researchers and developers can learn more about how people and AI interact on a psychological level and make systems that can adapt to users' emotions and suit their needs. Today, we are experiencing the so-called era of chatbot explosion [9]. The remarkable advancements in technologies, particularly artificial intelligence (AI), have significantly increased the adoption of digital conversational agents by enterprises. Chatbots are capable of simultaneously and immediately responding to multiple user messages in real-time [6]. Chatbots today have come a long way in comprehending languages and producing proper and useful answers [11]. Artificial intelligence and large language models (LLMs) have come a long way in a short amount of time. This has fundamentally revolutionized how companies today support their customers. Digital conversational agents today, considered to be a modern and strategic businesses tool. AI chatbots are useful because they are available 24/7, save money on running costs, and can answer a lot of user questions straight away [6]. Nonetheless, a critical question arises concerning the human dimension of customer service: Can chatbots emotionally engage users and exhibit artificial empathy? AI-powered chatbots possess the capability to express emotions and have achieved substantial popularity in customer service contexts [4]. As a result, there has been a huge increase in interest in AI software from academics, professionals, and organizations. This is especially true since well-known models, for instance ChatGPT, which is now in its fourth version, GPT-4, can do many tasks, like automatization and generative pictures or videos. Chatbots could change the way businesses service by connecting across different industries [9] by giving businesses a competitive edge [12]. As chatbots are getting better today at mimicking how human beings interact with each other, it would be critical to analyze how users react to this technology. Concerning empathy, AI has advanced to a point where users may no longer distinguish between genuine and artificial emotions [2]. Recent research indicates that any form of empathy can generate positive outcomes, including fostering trust [5]. Another study also proves that chatbots can show as much

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empathy as human teacher [1]. [5] states that empathy is one of the most important elements for developing trust and happiness, making the overall customer service experience better. Particularly within digital environments lacking physical interaction, empathy can act as a humanizing tool, enhancing perceived service quality [4]. According to recent studies, chatbots exhibiting high levels of emotional intelligence may elicit feelings of trust and satisfaction among users comparable to those elicited by human employees [1]. Nevertheless, artificial empathy remains a complex and controversial field. Although technology enhances AI chatbot's ability to mimic emotional expressions, many researchers are investigating whether such expressions are authentic and if users can perceive this level of empathy as mere "illusions" [2]. The genuine level of emotional responses directly influences perceptions of anthropomorphism, ultimately affecting the acceptance of chatbots as viable alternatives for customer service.

2 RESEARCH QUESTIONS & OBJECTIVES

This study explores the user perceptions between four variables: service quality, trust, emotional engagement, and satisfaction. There is a difference in the perception of empathy provided by these two AI models. The selection of ChatGPT and Pi AI was based firstly on how different their design is and secondly the way they communicate at their level of empathy. Open AI's ChatGPT is a more neutral and task-oriented AI model, while Pi AI is largely about emotional intelligence and being empathetic. A lot of research has employed Pi AI, mostly in schools, and it seems to always work out well [3]. Because they use different methods to express empathy, it is useful to do this comparison. This difference level of empathy, provides useful information on how users see, feel, and react to different kinds of AI powered chatbots in their customer service. Second, by comparing different types of artificial empathy, this study tries to enhance the understanding by giving information to competent programmers of how AI chatbots could be designed be in the future to be more effective. The research also studies how anthropomorphism and emotional artificial intelligence could change how digital customer services are delivered. In conclusion, this study, by comparing two models, also examine, how the empathy of these two chatbots, can improve user's views of service quality and help build long-lasting relationships based on trust. The study also looks at whether users can tell when someone provides low level of empathy and whether this affects how they feel about service quality, trust, and overall satisfaction.

RQ1: Does perceived empathy impact service quality, trust, and satisfaction?

RQ2: Are there measurable differences between ChatGPT and Pi AI in conveying empathy?

RQ3: Can users recognize and perceive artificial empathy during interactions with AI chatbots?

RQ4: How does the perceived artificial empathy of a chatbot influence users' cognitive evaluations (e.g., trust, satisfaction) and emotional responses?

RQ5: Do users value expressions of care and understanding from AI chatbots in a manner similar to human interactions, or do they dismiss them due to their artificial and non-human nature?

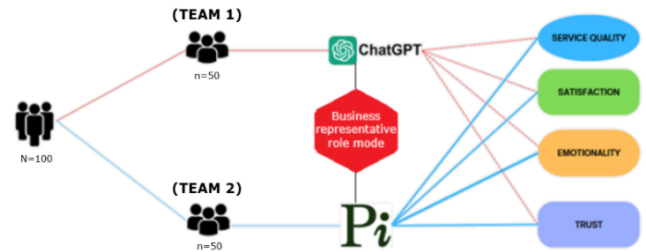


Figure 1: The overall experimental process

RQ6: Can the expression of artificial empathy by an AI chatbot serve as a significant determinant in enhancing users' overall perception of service quality?

Considering that expressions of care and understanding are valued in everyday interpersonal relationships, a critical question emerges: do users similarly value these expressions from an AI chatbot, or do they reject them due to their technical and unconventional nature? The research question specifically focuses on the impact of artificial empathy as a determinant of customers' perceptions of service quality. Can software, through the expression of sympathy and understanding, effectively enhance user perceptions of service quality? Conclusively, this research seeks to contribute to the understanding of this multifaceted relationship and to address the research questions.

3 METHODOLOGY

At this stage, the experimental design has been finalized, and all interactions with the artificial intelligence chatbots have been conducted. In this research, a comparative analysis between two AI software applications, ChatGPT and Pi AI, will be conducted. Compared to ChatGPT, Pi AI represents a contemporary AI chatbot explicitly designed to strongly convey empathy to users. An experimental will be used. Participants will first be invited to interact with both AI chatbots in a set situation. In this case, sample of the experiment, will pretend to be customers who are having hypothetical issues with their customer service or a product. There will be 100 people in the sample (n=100), and they will be split into two groups of 50 (n=50). To get more accurate results, each group will only interact one chatbot. This will ensure that each group's experience is unique by receiving different level of empathy. The first group (TEAM 1) will exclusively interact with ChatGPT, whereas the second group (TEAM 2) will interact solely with Pi AI. To address their issues, participants, hypothetically unable to successfully communicate with the e-shop regarding their problem—will resort exclusively to online service through the chatbot.

At the beginning of their experience with each chatbot, participants will receive instructions on how to get each chatbot to act like a real customer support agent (customer agent mode) by giving a specific command to the chatbot. This ensures, that each software, will be able to give more specific responses like a real human customer support agent. Participants will be given 15 pre-set questions through a questionnaire to rate their experience.

Table 1: The research variables

Variables	
Service Quality	Satisfaction
Emotionality	Trust

To record the evaluation of the sample, a Likert scale consisting of 32 questions will be utilized. Specifically, eight questions will be assigned to each of the four variables under investigation. The questions will be rated on a 5-point scale, ranging from 1 to 5. This scale will allow participants to express varying degrees of agreement/disagreement or satisfaction/dissatisfaction.

4 RESULTS OF THE STUDY

According to the results, 66% of the participants were men and 34% were women. Most of them (56%) were under 35 years old. Users between ages of 35 and 60 made up 34% of the group, while people beyond 60 made up only 10%. In terms of education, 44% of the people who took part had a bachelor's degree, 30% had a master's degree, and a lesser fraction had either a high school diploma or a PhD degree. Also, 44% of the people who took part had used AI software before, whereas 26% had never used any other AI application before.

4.1 SERVICE QUALITY

Results about service quality proves that the participants of the first sample did not believe that ChatGPT is able to provide service quality similar to a human. Only 4% of participants claimed that service was good as human representative. In contrast, most participants who interacted with the Pi AI software (56%) concluded that the software offered a positive score (4/5) was observed in 48% of responses level of service quality that resembled human interaction. Moreover, 20% of participants reported a highly appropriate experience, describing the service quality as fully comparable to that provided by a human customer service agent. They were also asked to express whether the two AI models gave them recommendations during the service and whether this changed their perception of service quality. Users claimed that ChatGPT was not that good in this variable because only 6% of the sample said they got advice during their experience. On the other hand, Pi AI seems to be more effective, as 58% of participants got helped and be given advises by the software. Another element that participants observed while judging each AI chatbot was if each model could give precise and personalized information throughout the customer service experience. When users interacted with ChatGPT, only 6% of them thought they got answers that were specific useful to them. In fact, 64% of the participants had mostly negative opinions, giving the software a low or moderate grade (2/5 or 3/5 on the Likert scale) for its capacity to give individualized information. People in the second group (TEAM 2) seem to be happier with Pi AI 40% of them agreed that the answers they got were focused and personalized enough. An additional critical factor influencing the perceived quality of service was the ability of the AI software to resolve every query raised by the user during the interaction. ChatGPT was not regarded as

**Figure 2: Results of positive answers on each variable**

fully capable in this aspect, with only 8% of the total sample indicating that the chatbot successfully addressed all their questions. A larger proportion 38% rated its performance in resolving queries as moderate. In contrast, the feedback from participants who interacted with Pi AI was significantly more favorable, with 54% of respondents reporting that the software was able to address all their inquiries to a very acceptable extent. When asked to evaluate whether the customer service provided by each chatbot was simple and easy to understand, 12% of participants who interacted with ChatGPT responded positively and expressed complete satisfaction. In comparison, 30% of participants in the Pi AI sample stated that they were fully satisfied, indicating that the service provided was clear and easily comprehensible. 20% of TEAM 1, found the service of ChatGPT satisfied in terms of immediacy and reaction time. On the other hand, 18% of Pi AI users claimed that service experience was worse because of small delays in reaction time. It was observed that participants who interacted with ChatGPT reported a positive emotional response to the quality and manner of service provision at a rate of only 6%. In contrast, double that percentage 12% of the total sample expressed a higher level of enthusiasm when interacting with the Pi AI software. In comparison, among participants who received service through the Pi AI software, 10% believed that the system had sufficient knowledge to effectively assist a customer, while an additional 33% expressed this belief to a highly convincing degree.

4.2 TRUST

Pi AI, elicited significantly higher levels of trust compared to ChatGPT, which was perceived as generating only moderate to low levels of trust. This suggests a lower overall perception of trustworthiness associated with ChatGPT among the participants. Specifically, 60% of the 50 participants reported experiencing a sense of trust toward Pi AI, whereas only 14% expressed similar sentiments in relation to ChatGPT. Moreover, only 12% of participants indicated that they would trust Pi AI for future customer service interactions, while this percentage dropped to 8% for ChatGPT. According to the findings, Pi AI inspired considerably higher levels of trust, with 60% of respondents rating their trust at a highly adequate level, compared to just 14% for ChatGPT. Participants reported feeling that Pi AI was more reliable and inspired a stronger sense of confidence and trust. These figures are further illustrated in the data

Table 2: Comparative analysis of positive and negative responses for ChatGPT and Pi AI across evaluation criteria

Variable	ChatGPT Positive responses)	Pi AI Positive responses	Percentage Difference	ChatGPT Negative responses	Pi AI Negative responses	Percentage Difference
Emotionality	118	268	+127,12%	123	49	−60,16%
Satisfaction	79	256	+224,05%	176	42	−76,14%
Service Quality	90	288	+220,00%	186	60	−67,74%
Trust	79	234	+196,20%	173	65	−62,43%

presented in the table below. Participants from Group 1 (TEAM 1) reported absolute trust in ChatGPT at a rate of just 6%. In contrast, in response to the same question regarding Pi AI, 12% of the participants from Group 2 indicated that they trusted the software to a fully rewarding degree. Only 8% of participants considered the responses provided by ChatGPT during the service interaction to be fully reliable and capable of instilling a sense of trust as customers. Simultaneously, 42% perceived the reliability of its responses to be at a moderate level. Conversely, 48% of participants in Group 2 reported that the responses of the Pi AI were sufficiently reliable to appropriate degree, which made them feel a sense of trust. Furthermore, 14% of participants rated the reliability of Pi AI's responses as fully effective in terms of fostering trust during the interaction. A particularly significant finding is that only 8% of participants would fully trust a similar AI-based software such as ChatGPT for future customer service interactions. Additionally, only 10% felt that the responses provided by ChatGPT were prompt, which in turn contributed to their sense of trust in the system. In contrast, 12% of participants in the second group stated that they would trust the Pi AI software for future service interactions. Notably, 46% believed that the responses provided by Pi AI were prompt, and this responsiveness was positively associated with perceived trust, as measured by the Likert scale. According to the findings, ChatGPT was considered capable of generating an overall sense of trust in a customer facing a problem like the one presented in the experiment by only 8% of participants in the first group. Additionally, 38% of respondents indicated that they perceived this capability at a moderate or slightly effective level. In contrast, within the second group, 16% of participants evaluated the Pi AI software as fully capable of instilling a strong sense of trust in the context of a similar customer service issue. The vast majority 52% of participants reported no sense of concern regarding potential misuse of personal data by the software. This exact percentage was mirrored in Group 2, with 52% of participants expressing the same response regarding the Pi AI software. Only 6% of the sample believed that ChatGPT provided service free from potential bias, whereas 18% of participants reported that Pi AI did not exhibit any signs of biased behavior during the interaction. Finally, only 8% of participants in Group 1 considered ChatGPT capable of building trust-based relationships in the role of a customer service representative. In Group 2, merely 4% of participants believed that Pi AI could establish such trust-based relationships.

4.3 EMPATHY

First group indicated that only 6% of participants believed that ChatGPT demonstrated clear signs of empathy during the customer service interaction. Conversely, 18% of respondents stated that the software exhibited no elements of empathy whatsoever. In contrast, among participants in the second group, 14% fully agreed that the Pi AI software displayed a strong sense of empathy during the service experience. The provision of empathy and emotional responsiveness to the hypothetical problems presented in the experiment was rated as fully positive by only 6% of participants in the first group who interacted with ChatGPT, while the majority (50%) perceived this provision as moderately. In contrast, 20% of participants in the second group considered the Pi AI software to have demonstrated empathy to a fully pleasant degree, while the majority 48% felt that empathy was provided to a fairly decent extent. It is noteworthy that 20% of the second group, which interacted with Pi AI, strongly believed that the AI software genuinely cared about their hypothetical problem. This contrasts with the first group, where only 6% believed that ChatGPT showed genuine emotional understanding and interest in their issue. An important aspect of emotionality is the perceived emotional responsiveness of the system. Within the first group, only 8% of participants considered ChatGPT's emotional response to be similar to that of a human to a fully satisfying degree. Additionally, only 4% felt that the software provided empathetic service comparable to that of a real human employee. In contrast, 22% of participants in the second group believed that Pi AI demonstrated an emotional response like a human, while 18% felt that they were served with empathy resembling that of an actual employee. Finally, 28% of participants in the first group believed that the hypothetical use of emoticons by ChatGPT would have further negatively affected their perception of its empathy. In comparison, 18% of the second group expressed the same opinion regarding Pi AI. Only 6% of the first group responded positively in relation to ChatGPT, whereas 20% of the second group stated that empathy provided by Pi AI significantly improved their experience.

4.4 SATISFACTION

The first sample, which interacted with ChatGPT, reported that the software met their expectations to some extent. However, only 6% stated that they were fully satisfied with the experience and found it easy to feel a sense of satisfaction. In contrast, participants in the second group, who were served via Pi AI, reported a significantly higher level of satisfaction, with 24% declaring they were fully satisfied with the service provided by the software. This difference proves that Pi AI was perceived as more capable of meeting users'

expectations and delivering a satisfied customer experience. The promptness of the software's responses, as a factor contributing to the sense of customer satisfaction, was considered a realistic and meaningful aspect by 12% of participants in the first group who interacted with ChatGPT. In contrast, nearly double that percentage 22% of the second group expressed the same opinion regarding Pi AI. Only 6% of participants in the ChatGPT group stated that the level of satisfaction provided by the software was fully sufficient for them to choose it again in the future, while 42% held this view to a moderately adequate extent. Participants in the Pi AI group, on the other hand, indicated that the level of satisfaction was fully adequate and that they would choose to be served by the software again in the future at a rate of 24%. Politeness expressed by the AI software was also identified as a contributing factor to customer satisfaction during the evaluation process. In the first sample, 38% of participants felt satisfied with ChatGPT, while 14% reported they did not received politeness at all. On the other side, 54% of participants in the second sample reported perceiving politeness and satisfaction by Pi AI. Furthermore, ChatGPT was perceived to provide satisfaction comparable to that of a human representative to a slightly sufficient extent by 42% of the participants, with 14% simultaneously expressing a negative view regarding this comparison. In contrast, 52% of the participants who interacted with Pi AI felt that the satisfaction provided by the software was comparable to that of a human representative. ChatGPT was considered capable of delivering personalized information by only 4% of participants in the sample. Similarly, a small proportion of the second group those who interacted with the Pi AI software, reported a comparable experience, with only 10% believing that the software provided personalized information. Finally, regarding clarity of responses, 6% of participants in the first group stated that ChatGPT's answers were clear and contributed to their satisfaction. In comparison, participants in the second group reported at a significantly higher rate 22% that the responses provided by Pi AI were clear and offered a satisfactory experience.

5 CONCLUSION AND FUTURE WORK

This research examined user's perception about trust, satisfaction and service quality when they interact with an empathetic or a non-empathetic AI chatbot into the role of human representative during digital customer service. The results of the experiment, proved that presence of high level of artificial empathy in a chatbot, seem to affect the user perception about the four variables of this study. The results, also prove, that users found Pi AI to be more emotionally expressive and empathetic providing personalized suggestions and advice. Users claimed that ChatGPT seem to be more organized and effective in some factors like time responses, but it was less emotionally engaging. Although, sample still do not fully embrace or trust AI chatbots. In the future, researchers could investigate if long-term user interactions with empathic chatbots could affect their perception of trust, satisfaction and service quality.

Satisfaction and service quality variable had the biggest drop in user dissatisfaction. Pi AI had more than 75% more positive responses and 67% less negative comments than ChatGPT. These results back up the premise that making AI chatbots more emotionally responsive not only makes users happier, but also lessens

the effects of bad experiences. The study's results also give us vital information about how to make AI chatbots that can understand and respond to people's feelings. Pi AI's scores in trust, satisfaction, and perceived empathy suggest that adding emotionally smart cues like context-aware affirmations, flexible tone, and personalized recommendations can make the whole user experience much better. Designers should put cognitive empathy mechanisms as priority. This would let chatbots to identify in real time whenever a user is upset or worried and reply in a socially appropriate and emotionally attuned way. The results also show how important it is to avoid showing emotions in a shallow or overdone way, which consumers can see as non-real. So, future chatbot design should aim for emotional responsiveness that is balanced and based on clear, ethical,

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